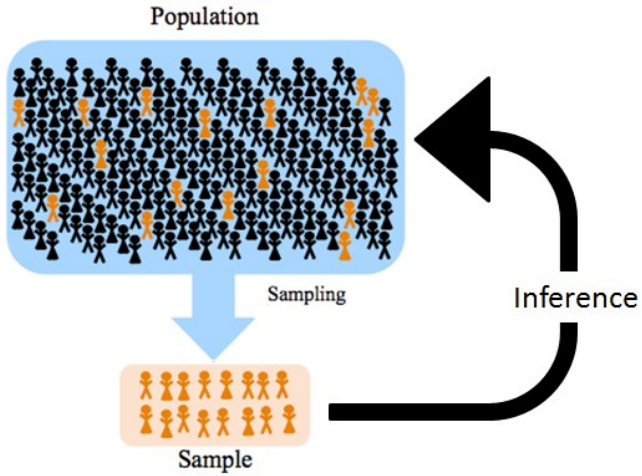
Lecture 21

Statistical Inference I



Announcements

- Course project deliverable 1 (exploration) due tonight @ 11:59pm
  - Guided questions up later this week

- Office hours every Tuesday (noon to 4pm) and Friday (11am to 1pm)
  - Link to book appointments is on the course website

Last Time

How to effectively report your findings

- Technical tools: Markdown, LaTeX (math mode), HTML
- Style: breaking report into sections, using headings, etc.
- Strategies: discuss results in the introduction, label tables and plots and refer to them in the text, etc.

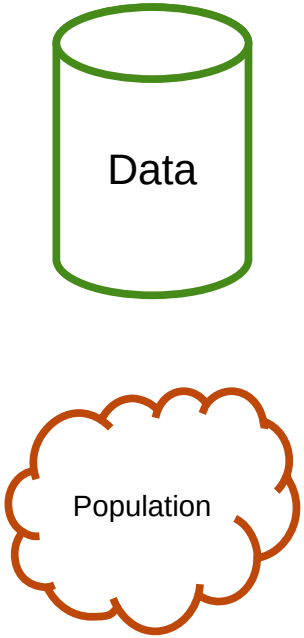
Today

Statistical Inference

- Sampling
- Sample Variance
- Statistical Hypotheses
- Difference of Means Test

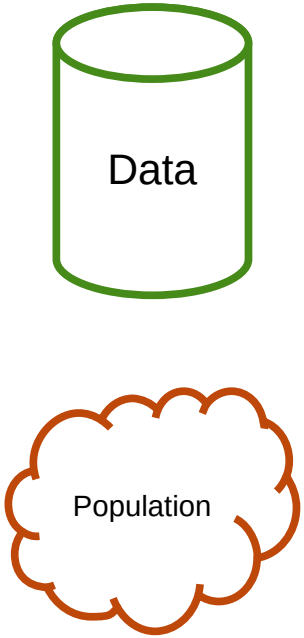
Samples & Populations

- For most of this course, we've worked "inside the box"
  - Focused on analyzing a dataset
  - Today, we're going to start reasoning about the population
- Think of our dataset as a **sample** from the population
  - A sample is often a concrete dataset
  - The population is often an idealized category



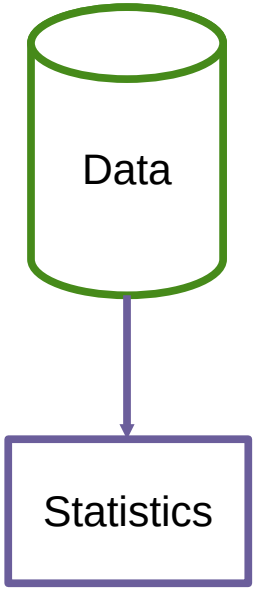
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- Example
  - Sample: wait times at Meadville Starbucks collected today
  - Wait times during the morning at all Starbucks



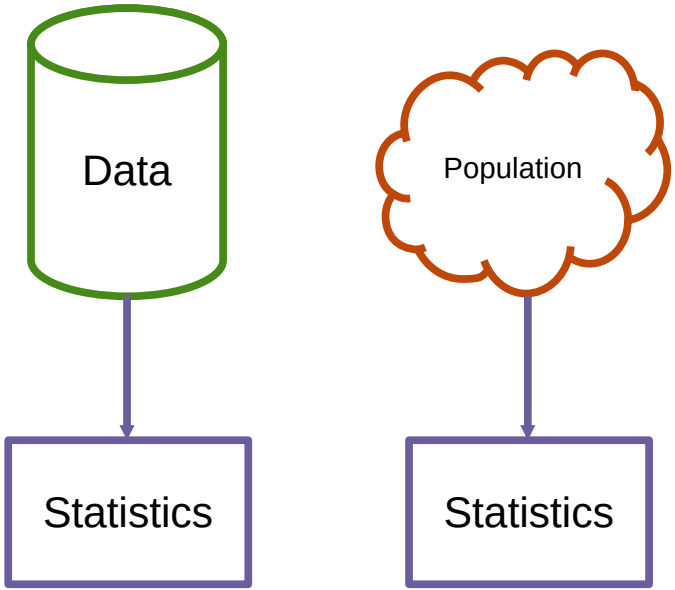
Statistics

- Summaries of our data
- We can learn a lot about a dataset through its statistics



Statistics

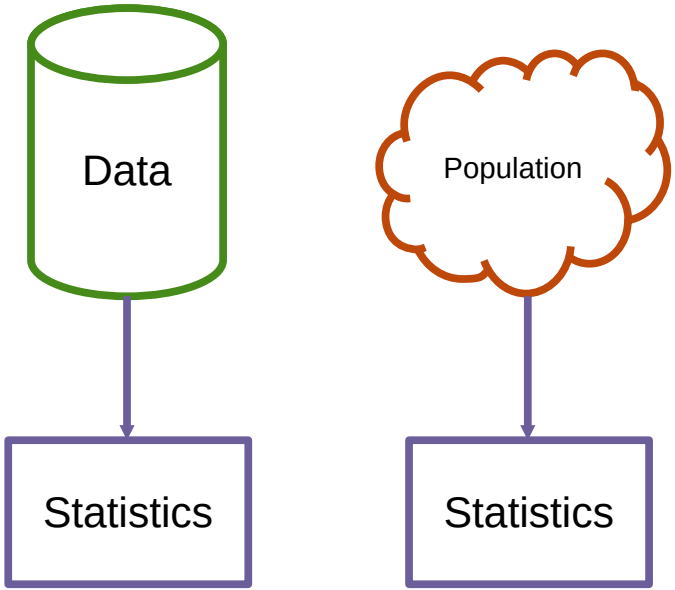
- Summaries of our data
- We can learn a lot about a dataset through its statistics
- Populations have statistics too
- We can sometimes learn about the population statistics from our sample statistics
  - Called **inference**



Representative Samples

- A sample is **representative** of the population if its statistics match those of the population

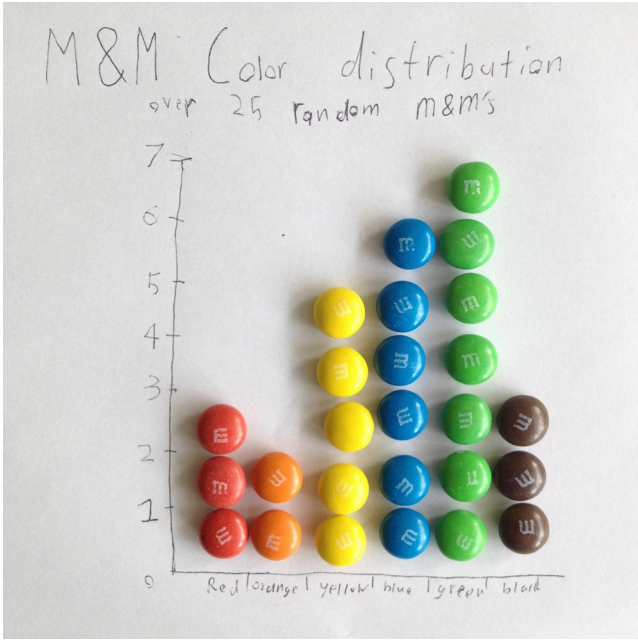
- A representative sample can be difficult to obtain!



Sample Variation

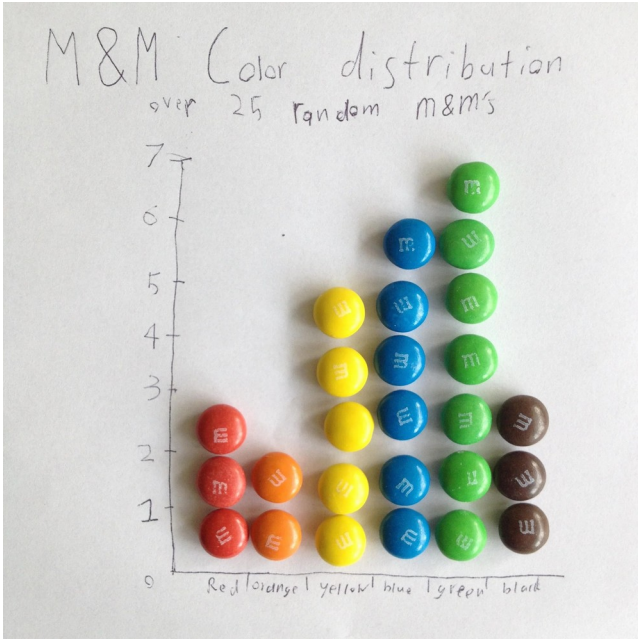
- When you open a bag of M&Ms, do you get the exact same ratios every time?

- Even if the population ratio is uniform, under sampling, it's reasonable to expect some variation



Sample Variation

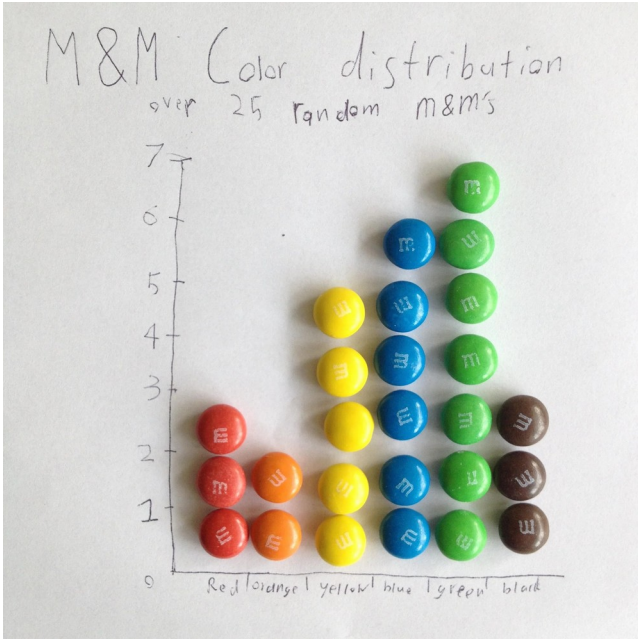
- When you open a bag of M&Ms, do you get the exact same ratios every time?
- Even if the population ratio is uniform, under sampling, it's reasonable to expect some variation
- What do you predict would happen if we open lots of bags of M&Ms and looked at the color ratios?



Sample Variation

- We say a sampling procedure is **unbiased** if, on average, the sample statistics match the population statistics

- Random sampling is unbiased!



Sampling Variation

- Let's look at the demo

Statistical Hypotheses

Ask: does a relationship in the sample hold for the population?

- Assume our sample was randomly sampled



Statistical Hypotheses

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- Not necessarily, the observed relationship may be due to sampling variance



Statistical Hypotheses

Ask: does a relationship in the sample hold for the population?

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Answer this by measuring whether or not the relationship differs from chance

Next Time

- Testing statistical hypotheses